YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 2 SOLUTIONS

Combinatorics: Pascal's triangle

1 a Each number is the sum of the two numbers above it. **[2]**

b

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

1 7 21 35 35 21 7 1

c $x^2 + 2ax + a^2, x^3 + 3ax^2 + 3a^2x + a^3, x^4 + 4ax^3 + 6a^2x^2 + 4a^3x + a^4,$

$x^5 + 5ax^4 + 10a^2x^3 + 10a^3x^2 + 5a^4x + a^5$ **[2]**

The coefficients are the numbers in that row (+ 1) of Pascal's triangle.

(e.g. coefficients of $(x + a)^4$ are in row 5.) Students should be rewarded for calling the first 1 row 0, corresponding to $(x + a)^0$. **[1]**

Using **b**, $x^7 + 7ax^6 + 21a^2x^5 + 35a^3x^4 + 35a^4x^3 + 21a^5x^2 + 7a^6x + a^7$ **[1]**

d The number of ways to select 4 xs out of 7, ${}^7C_4$. **[1]**

e The number of ways to select r xs out of n, ${}^nC_r$. **[1]**

f Consider $(x + a)^{n+1} = (x + a)(x + a)^n$

$= (x + a)(x^n + \ldots + Aa^{n-r}x^{r+1} + Ba^{n-r+1}x^r + \ldots + a^n)$ **[1]**

$a^{n-r}x^{r+1}$ term $= a \times Aa^{n-r}x^{r+1} + x \times Ba^{n-r+1}x^r$ **[1]**

$= Aa^{n-r+1}x^{r+1} + Ba^{n-r+1}x^{r+1}$

$= (A + B)a^{n-r+1}x^{r+1}$ **[1]**

This confirms that the adding rule for Pascal's triangle also works for the expansion of $(x + a)^n$. **[1]**

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g

$${}^0C_0$$

$${}^1C_0 \quad {}^1C_1$$

$${}^2C_0 \quad {}^2C_1 \quad {}^2C_2$$

$${}^3C_0 \quad {}^3C_1 \quad {}^3C_2 \quad {}^3C_3$$

$${}^4C_0 \quad {}^4C_1 \quad {}^4C_2 \quad {}^4C_3 \quad {}^4C_4$$

$${}^5C_0 \quad {}^5C_1 \quad {}^5C_2 \quad {}^5C_3 \quad {}^5C_4 \quad {}^5C_5$$

$${}^5C_0x^5 + {}^5C_1ax^4 + {}^5C_2a^2x^3 + {}^5C_3a^3x^2 + {}^5C_4a^4x + {}^5C_5a^5$$

$${}^nC_0x^n, {}^nC_1ax^{n-1}, {}^nC_2a^2x^{n-2}, {}^nC_3a^3x^{n-3}$$

${}^0C_0 = 1$ since there is one way of selecting no objects from the null (empty) set.

$$\frac{0!}{0!(0-0)!} = 1 \text{ as } 0! = 1$$ **[2]**

h i You can choose 0 objects in ways, or 1 object in ways, or 2 objects in ways...

Number of ways $= {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n$. **[2]**

ii $(x+a)^n = {}^nC_0x^n + {}^nC_1ax^{n-1} + {}^nC_2a^2x^{n-2} + {}^nC_3a^3x^{n-3} + \ldots + {}^nC_na^n$ **[1]**

$(1+1)^n = {}^nC_01^n + {}^nC_11\times1^{n-1} + {}^nC_21^2\times1^{n-2} + {}^nC_31^3\times1^{n-3} + \ldots + {}^nC_n1^n$ **[1]**

$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n$

${}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n$ **[1]**

iii e.g. $1+3+3+1 = 8 = 2^3$, $1+4+6+4+1 = 16 = 2^4$ **[1]**

iv For each object, to choose or not to choose is 2 possibilities.

Over n objects, this is $2\times2\times2\times\ldots$ (n times) $= 2^n$. **[1]**

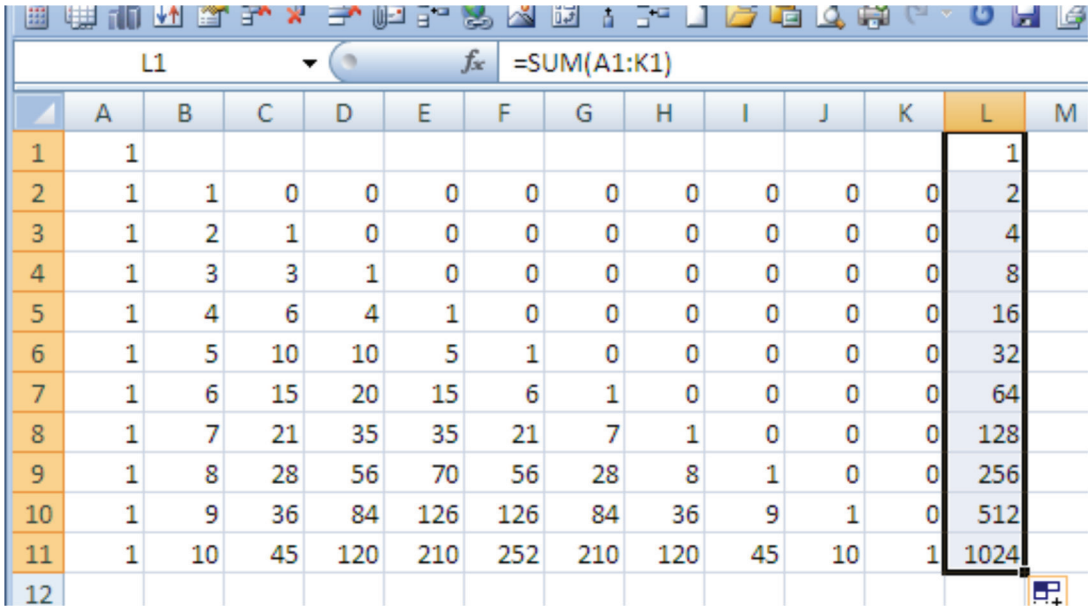
This is also the number of ways of choosing 0 objects plus the number of ways of choosing 1 object plus the number of ways of choosing 2 objects etc. so

$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n$ as before. **[1]**

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2 a The spreadsheet is below. **[5]**

b The sums should be powers of 2. **[2]**



L1 | =SUM(A1:K1)

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	1											1	
2	1	1	0	0	0	0	0	0	0	0	0	2	
3	1	2	1	0	0	0	0	0	0	0	0	4	
4	1	3	3	1	0	0	0	0	0	0	0	8	
5	1	4	6	4	1	0	0	0	0	0	0	16	
6	1	5	10	10	5	1	0	0	0	0	0	32	
7	1	6	15	20	15	6	1	0	0	0	0	64	
8	1	7	21	35	35	21	7	1	0	0	0	128	
9	1	8	28	56	70	56	28	8	1	0	0	256	
10	1	9	36	84	126	126	84	36	9	1	0	512	
11	1	10	45	120	210	252	210	120	45	10	1	1024	
12													

c The triangular numbers are found in the diagonal parallel to 1, 1, 1... but starting 1, 3, 6, 10…. the numbers are third from the left or third from the right.

They appear in two diagonal rows.

In the spreadsheet they are column C or the diagonal A3, B4, C5...

They are ${}^{2}C_{0}$, ${}^{3}C_{1}$, ${}^{4}C_{2}$, ${}^{5}C_{3}$ ... **[2]**

The rth triangular number is ${}^{r+1}C_{r-1}$ (from Pascal's triangle with combinations). **[1]**

[3]

$$
\begin{aligned}
{}^{r+1}C_{r-1} &= \frac{(r+1)!}{(r-1)!((r+1)-(r-1))!} \\
&= \frac{1 \times 2 \times 3 \times \ldots \times (r-1) \times r \times (r+1)}{1 \times 2 \times 3 \times \ldots \times (r-1) \times 2!} \\
&= \frac{r(r+1)}{2}
\end{aligned}
$$

d The result is a square number. $(1 + 3 = 4, 3 + 6 = 9$, etc.) **[1]**

The $(r+1)^{\text{th}}$ triangle number is $\frac{(r+1)(r+2)}{2}$ **[1]**

$$
\begin{aligned}
\frac{r(r+1)}{2} + \frac{(r+1)(r+2)}{2} &= \frac{(r+1)(r+r+2)}{2} \\
&= \frac{(r+1)(2r+2)}{2} \\
&= (r+1)^2
\end{aligned}
$$

[2]

OR

$$\frac{r(r+1)}{2} + \frac{(r+1)(r+2)}{2} = \frac{r^2 + r + r^2 + 3r + 2}{2}$$
$$= \frac{2r^2 + 4r + 2}{2}$$
$$= \frac{2(r^2 + 2r + 1)}{2}$$
$$= (r+1)^2$$

[2]

On the spreadsheet, enter a suitable sum, ideally the first in an empty square.

e.g. in cell C13, enter = C3+C4 (giving 4)

If this is dragged down, the next cell contains '=C4+C5' giving 9, and so on.

Teachers can decide how far students should go. **[2]**

e Teachers to check enlarged Pascal's triangle.

$^{14}C_9 = 2002$ and it is in cell J15. (Remember the row starts with $^{14}C_0$.) **[3]**

f Either a diagonal line with the 'sum' cell vertically underneath the last cell, or a column with the 'sum' cell diagonal to the last cell. Teacher to check verification. An example is shown below.

C18 f_x =A8+B9+C10

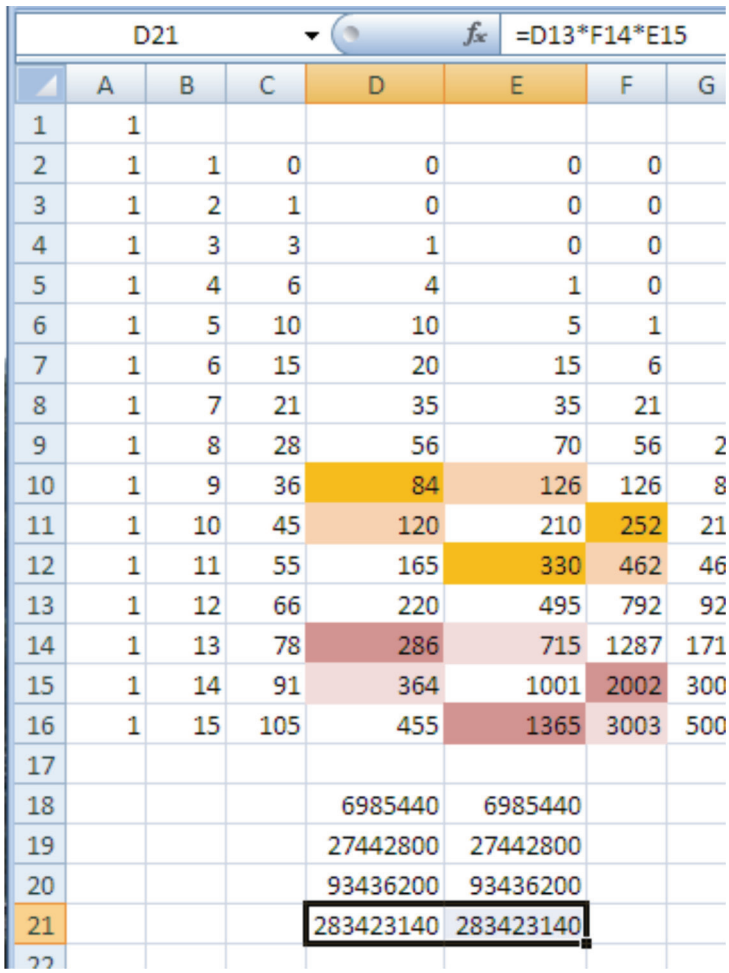
	A	B	C	D	E	F	G	H	I
1	1								
2	1	1	0	0	0	0	0	0	
3	1	2	1	0	0	0	0	0	
4	1	3	3	1	0	0	0	0	
5	1	4	6	4	1	0	0	0	
6	1	5	10	10	5	1	0	0	
7	1	6	15	20	15	6	1	0	
8	1	7	21	35	35	21	7	1	
9	1	8	28	56	70	56	28	8	
10	1	9	36	84	126	126	84	36	
11	1	10	45	120	210	252	210	120	4
12	1	11	55	165	330	462	462	330	16
13	1	12	66	220	495	792	924	792	49
14	1	13	78	286	715	1287	1716	1716	128
15	1	14	91	364	1001	2002	3003	3432	300
16	1	15	105	455	1365	3003	5005	6435	643
17									
18			45			1716			

Note: C18 =A8+B9+C10 − It equals 45, the same as C11.

F18: =SUM(F6:F13) − It equals 1716, the same as G14. **[5]**

g There are a variety of different descriptions possible, e.g. a 3×3 square with the 'rising' diagonal missing, two 3-unit L shapes as opposite corners of a 3×3 square, or other possibilities. You are still multiplying every second marked number as you go round the perimeter of the 3×3 square. The teacher can check other possible descriptions and also the individual verifications. Most descriptions would be correct. An example is shown below. Students may also wish to compare the shapes from the two types of grids in parts **f** and **g**.

The drag down works as it produces a new pattern with the same rule.



D21 f_x =D13*F14*E15

	A	B	C	D	E	F	G
1	1						
2	1	1	0	0	0	0	
3	1	2	1	0	0	0	
4	1	3	3	1	0	0	
5	1	4	6	4	1	0	
6	1	5	10	10	5	1	
7	1	6	15	20	15	6	
8	1	7	21	35	35	21	
9	1	8	28	56	70	56	2
10	1	9	36	84	126	126	8
11	1	10	45	120	210	252	21
12	1	11	55	165	330	462	46
13	1	12	66	220	495	792	92
14	1	13	78	286	715	1287	171
15	1	14	91	364	1001	2002	300
16	1	15	105	455	1365	3003	500
17							
18				6985440	6985440		
19				27442800	27442800		
20				93436200	93436200		
21				283423140	283423140		

The lightly coloured numbers are multiplied together and the darker coloured numbers are multiplied together.

Dragging down produces new sets of numbers. The last one is the set of red numbers, so you can go no further without extending the triangle. **[5]**

3 a The number of ways of choosing r objects from n objects to be in a selection is the same as the number of ways of choosing $(n - r)$ objects out of n objects to not be in the selection. **[1]**

$$^nC_r = \frac{n!}{r!(n-r)!}$$

$$^nC_{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!}$$ **[2]**

$$^nC_r = {}^nC_{n-r}$$

b Odd terms $= {}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots$ **[1]**

If there are an even number of terms, ${}^nC_n$ will be an even term. **[1]**

Even terms $= {}^nC_n + {}^nC_{n-2} + {}^nC_{n-4} + \ldots = {}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots =$ even terms since ${}^nC_r = {}^nC_{n-r}$. **[2]**

c The sum of the odd terms is still equal to the sum of the even terms. **[1]**

Note to teacher: The proof (I suspect) is beyond most students at this level.

d The sum of the terms is a power of two, and for this to happen it would have to be a multiple of 3. **[1]**

e The situation is shown below. It's almost as if they do their best to be equal. There is a repeating cycle which repeats after every six sets with two sums equal and the third either one more or one less than the other two.

This is a purposefully open-ended question. Answers are only wrong if the pattern described does not fit the numbers. It is suggested 1 mark for every different thing students observe about the numbers, up to 3 marks maximum. Teachers should use their discretion. **[3]**

D18 f_x =A3+D3+G3+J3+M3+P3

	A	B	C	D	E	F	G
1	1						
2	1	1	0	0	0	0	0
3	1	2	1	0	0	0	0
4	1	3	3	1	0	0	0
5	1	4	6	4	1	0	0
6	1	5	10	10	5	1	0
7	1	6	15	20	15	6	1
8	1	7	21	35	35	21	7
9	1	8	28	56	70	56	28
10	1	9	36	84	126	126	84
11	1	10	45	120	210	252	210
12	1	11	55	165	330	462	462
13	1	12	66	220	495	792	924
14	1	13	78	286	715	1287	1716
15	1	14	91	364	1001	2002	3003
16	1	15	105	455	1365	3003	5005
17							
18				1	2	1	
19				2	3	3	
20				5	5	6	
21				11	10	11	
22				22	21	21	
23				43	43	42	
24				85	86	85	
25				170	171	171	
26				341	341	342	
27				683	682	683	
28				1366	1365	1365	
29				2731	2731	2730	
30				5461	5462	5461	
31				10922	10923	10923	

Extra marks could be awarded at the teacher's discretion.